

OMAR REHAN

AI / ML Engineer · Computer Vision Specialist · Full-Stack Developer
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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Hugging Face](#) | [Kaggle](#) | [Medium](#) | [Tableau](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Full-Stack Developer with a B.Sc. in Computer Science and experience building machine learning, deep learning, and software solutions. Skilled in developing end-to-end systems, from data preparation and model training to deployment, integration, and full-stack web development. Experienced in delivering scalable, production-ready applications that combine AI with modern software engineering. Passionate about solving real-world problems through technology.

EDUCATION

Arab Academy for Science, Technology & Maritime Transport (AASTMT) Oct 2019 - Feb 2024
B.Sc. in Computer Science - College of Computing & Information Technology · Alexandria, Egypt

- Graduation Project: AI-Powered Smart Classroom Automation - TensorFlow Lite CNN on Raspberry Pi for real-time occupancy detection; Flutter/Bluetooth IoT control of lighting, HVAC & projectors (PIR, DHT11, LDR sensors).

PROJECTS & EXPERIENCE

SIVED: SAR Satellite Vehicle Detection | *YOLOv8-OBb · PyTorch · OpenCV · Gradio · Hugging Face* 2026

- 98.8% mAP@0.5** - Fine-tuned YOLOv8m-OBb on 12,013 annotated SAR images; SAR-specific augmentation (disabled HSV, applied intensity variation), converging in 70 epochs on Tesla T4.
- Deployed Gradio app on Hugging Face Spaces with Gaussian heatmaps, OBb visualization, and frame-by-frame video detection.

RehanPulse: Real-Time Developer Dashboard | *Next.js 14 · TypeScript · Firebase · Zustand · Three.js* 2026

- Full-stack hub: GitHub GraphQL API (HMAC-SHA256 webhooks), Vercel REST API & Firestore via SSE; AES-256-GCM token encryption, Playwright E2E tests, Vitest unit tests, Storybook component library.
- Pulse AI chatbot (Hugging Face Inference API, streaming SSE, live context injection); macOS frosted-glass UI with Three.js particle landing page and Framer Motion transitions.

Portfolio: Personal Website | *Next.js · React · TypeScript · Node.js · Supabase · Tailwind CSS · Vercel · Git/GitHub* 2026

- [Portfolio](#) | Additional deployed AI, Machine Learning, Computer Vision, and Full-Stack projects available online.

AstroVision: Llama 3.2-11B Vision Fine-Tuning | *Unsloth · LoRA/PEFT · TRL · PyTorch · Gradio* 2025

- Fine-tuned Meta's 11B multimodal LLM on 250 astronomy image-caption pairs with 4-bit quantization and LoRA (r=16, peak 10 GB GPU); evaluated with BLEU & ROUGE-L; full walkthrough published on Medium.

Brain Tumor MRI Classification | *InceptionV3 · TensorFlow/Keras · Grad-CAM · Scikit-learn* 2025

- End-to-end pipeline: InceptionV3 transfer learning, class-balanced augmentation, tf.data prefetching, Grad-CAM explainability, and multi-class AUC evaluation; deployed on Hugging Face Spaces.

IT Support & Data Management Specialist | *Egyptian Army · Electronic Warfare Dept · North Sinai* Apr 2024 - May 2025

- Managed digital records, provided IT & network support, and contributed to software development, automation, and AI integration initiatives.

TECHNICAL SKILLS

AI / ML & Vision: PyTorch, TensorFlow/Keras, YOLOv8-OBb, OpenCV, Hugging Face Transformers, Unsloth, LoRA/PEFT, LangChain, Grad-CAM, TFLite, Gradio, Scikit-learn
Full-Stack Dev: Next.js 14, React, TypeScript, Node.js, FastAPI, Django, Flask, PHP, PostgreSQL, MongoDB, Firebase, REST, GraphQL, Docker, AWS, Vercel
Languages & Tools: Python, JavaScript/TypeScript, SQL · GitHub, Google Colab, Raspberry Pi, IoT, Tailwind CSS, Three.js, Framer Motion

CERTIFICATIONS

AI Associate - Digital Transformation Program (IBM-based) - *Digital Hub* Jan 2023
IoT Associate - Digital Transformation Program (IBM-based) - *Digital Hub* Jan 2023
CCNAv7: Enterprise Networking, Security & Automation - *Cisco Networking Academy* Jul 2023
CCNAv7: Switching, Routing & Wireless Essentials - *Cisco Networking Academy* Feb 2023
CCNAv7: Introduction to Networks - *Cisco Networking Academy* Jun 2022

COMPETITIONS & ACTIVITIES

Hult Prize - Regional Finals Competitor (Cairo Impact Summit - UN-affiliated Global Entrepreneurship Competition) April 2021

LANGUAGES

Arabic - Native Proficiency **English** - Full Professional Proficiency